PS7: *All* the derivative shortcut rules!

Introduction: In this Problem Set, you'll show how to find derivatives **step-by-step**, carefully listing out all of your steps, and using good derivative grammar.

- I am going to be so annoying about making you take **very small steps**, because it's a good way to get comfortable with all the shortcut rules.
- I am also going to be so annoying about **good grammar**, because thinking carefully about what letter is the variable is important (and will be even more important next chapter).
- Lastly, I am going to be so annoying about **PEMDAS issues**, because it is important for us to understand that

  $$2y+4\cdot(\cos(y))$$

  is not the same as

  $$(2y+4)\cdot(\cos(y)),$$

  but it is even more important for us to understand **why**.

Instructions: For each problem you choose:

(a) Refer to the reference list of shortcut rules to identify **all** the derivative rules you're going to need to use for your problem.

- Watch out for the sneaky structure rules that you use without even noticing them!

(b) Make your sheet of paper into two columns.

- In the left column, do tiny derivative steps.
- In the right column, list all the rules you are using in that step.
- Look at the example on the back of the reference list page (the one we did as a warmup on Thursday) to see how tiny your steps should be.

(c) Repeat with a new problem until you have checked off **all 12 of the shortcut rules** in the reference list at least once.

- Yes, all three of the rules for trigonometric functions count separately.
- I am pretty sure you will need to do at least five problems.
- If you do extra problems, I will give you a sticker.

Throughout, label your work with derivative signs, using correct derivative notation and grammar.

Here are the problems you can choose from.
Remember that "differentiate" is the verb for "derivative." (You would think the verb would be "derive", but that actually means something else.)

(1) Find $\dfrac{dy}{dx}$ if $y = (2x-1)^7 - x + e$.

(2) If $L(y) = \tan^4(y) + \tan(4)$, find $L'(y)$. (Remember what $\tan^4(y)$ means?)

(3) Find $B'(x)$ if $B(x) = \cos(e^{5x})$.

(4) Differentiate $W(x) = \dfrac{x^2 - 5x + 11}{\sqrt{8x+7}}$.

(5) Differentiate $g(q) = \dfrac{q^3}{\ln\left(q^3 + \dfrac{1}{3}\right)}$.

(6) Differentiate $B(x) = (Ce^{4x} - D)(3x)$, where C and D are constants.

(7) Find $f'(y)$ if $f(y) = \left(\sqrt{y^3+1}\right)\left(\ln\left(\dfrac{y}{4}\right)\right)$. (Note: This doesn't require the quotient rule.)

(8) Find $\dfrac{dB}{dx}$ if $B = \dfrac{e^{2x+\pi}}{\sin(\pi x)}$.

(9) Differentiate $h(p) = \dfrac{(3p-1)^{1/4}}{\cos(2p-7)}$.

(10) Differentiate $f(q) = \cos(q)\sin\left(\dfrac{4}{\pi}\, q\right)$.

(11) Differentiate $G(x) = px^4 + qx^2 - x^r$, where $p,\ q$ and r are constants.

(12) Find the derivative of $V(x) = \left(5x - \dfrac{1}{7}\right)^{-10} - (2x)^{9/2}$.

(13) Find $\dfrac{dh}{dt}$ if $h(t) = 4^t\cos(t) + t^4\sin(t)$.

(14) Differentiate $h(t) = -5(2t^4 - 3)^9 + 7$.

(Example) Find $\dfrac{dp}{dt}$ if $p(t) = \dfrac{\cos(t^2-1)}{e^{-3t}}$.

I am going to need the following derivative rules:

- quotient rule
- chain rule
- sum rule
- constant multiple rule
- trig function rule (the one for $\cos(\ldots)$)
- power function rule
- constant function rule
- exponential function rule

Now, explain what rules are happening in each step, as highlighted by the blue boxes.

Steps **Rules**

$$\frac{dp}{dt} = \frac{[e^{-3t}]\cdot\frac{d}{dt}[\cos(t^2-1)] - [\cos(t^2-1)]\cdot\frac{d}{dt}[e^{-3t}]}{[e^{-3t}]^2}$$

$$= \frac{[e^{-3t}]\cdot\frac{d}{dt}[\cos(t^2-1)] - [\cos(t^2-1)]\cdot\boxed{[e^{-3t}\cdot\frac{d}{dt}[-3t]]}}{[e^{-3t}]^2}$$

$$= \frac{[e^{-3t}]\frac{d}{dt}[\cos(t^2-1)] - [\cos(t^2-1)]\cdot[e^{-3t}\cdot\boxed{(-3)}]}{[e^{-3t}]^2}$$

$$= \frac{[e^{-3t}]\cdot\boxed{\left[-\sin(t^2-1)\cdot\frac{d}{dt}(t^2-1)\right]} - [\cos(t^2-1)]\cdot[e^{-3t}\cdot(-3)]}{[e^{-3t}]^2}$$

$$= \frac{[e^{-3t}]\cdot[-\sin(t^2-1)\cdot\boxed{\left[\frac{d}{dt}(t^2)-\frac{d}{dt}(1)\right]}] - [\cos(t^2-1)]\cdot[e^{-3t}\cdot(-3)]}{[e^{-3t}]^2}$$

$$= \frac{[e^{-3t}]\cdot[-\sin(t^2-1)\cdot\boxed{(2t-0)}] - [\cos(t^2-1)]\cdot[e^{-3t}\cdot(-3)]}{[e^{-3t}]^2}$$

And now I am done, because I have no more "please take the derivative of" signs left. Yay!

Reference list of derivative shortcut rules

Here is the list of all the shortcut rules we've learned. Check them off as you use them. Keep this list for reference!

Structure rules

- ☐ sum rule: $\frac{d}{dx}[f(x)+g(x)] = \frac{d}{dx}[f(x)] + \frac{d}{dx}[g(x)]$
- ☐ constant multiple rule: $\frac{d}{dx}[c \cdot f(x)] = c \cdot \frac{d}{dx}[f(x)]$
- ☐ product rule: $\frac{d}{dx}[f(x) \cdot g(x)] = \frac{d}{dx}[f(x)] \cdot g(x) + \frac{d}{dx}[g(x)] \cdot f(x)$
- ☐ quotient rule: $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x) \cdot \frac{d}{dx}[f(x)] - f(x) \cdot \frac{d}{dx}[g(x)]}{[g(x)]^2}$
- ☐ chain rule: $[f(g(x))]' = f'(g(x)) \cdot g'(x)$, or in other words, $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot \frac{d}{dx}[g(x)]$

Function rules

- ☐ constant function rule: $\frac{d}{dx}[c] = 0$
- ☐ power function rule: $\frac{d}{dx}[x^n] = nx^{n-1}$
- ☐ exponential function rule: $\frac{d}{dx}[a^x] = a^x \cdot \ln(a)$

  trigonometric function rules:

  - ◯ $\frac{d}{dx}[\sin(x)] = \cos(x)$
  - ◯ $\frac{d}{dx}[\cos(x)] = -\sin(x)$
  - ◯ $\frac{d}{dx}[\tan(x)] = \sec^2(x)$

- ☐ logarithmic function rule: $\frac{d}{dx}[\ln(x)] = \frac{1}{x}$

Some extra function rules

These ones are not on this homework and generally come up less frequently, but I would get points on my math license if I didn't at least say them.

- inverse trigonometric function rules:
  - $\frac{d}{dx}[\arctan(x)] = \frac{1}{1+x^2}$
  - $\frac{d}{dx}[\arcsin(x)] = \frac{1}{\sqrt{1-x^2}}$
- rules for other trigonometric functions:
  - $\frac{d}{dx}[\cot(x)] = -\csc^2(x)$
  - $\frac{d}{dx}[\sec(x)] = \sec(x)\tan(x)$
  - $\frac{d}{dx}[\csc(x)] = -\csc(x)\cot(x)$